AI-powered Virtual Humans

LLM-Powered Cognitive Architectures

Module 1 · Chapter 01
Psychological Foundations
of Cognitive Architectures





Virtual Agents and Virtual Humans

From Agents to Virtual Agents

Agent

A system that perceives its environment and acts autonomously over time to pursue goals. [1]





Virtual Agent

An agent implemented in software. [2]

Using Tools

Agents can use tools to pursue their goals.

For virtual agents, tools are functions: software calling other software to achieve a goal.



From Virtual Agents to Virtual Humans



Virtual humans are a specialized class of virtual agents that mimic human presence and behaviour. [2]

- Human-like embodiment
- Perception and action
- Cognitive capabilities
- Social and emotional capabilities
- Verbal and non-verbal communication

Computational Cognitive Architectures

The Perceive–Reason–Act Loop



What Is Cognition?

The processes through which a system acquires, processes, and uses information to guide behaviour. [2]



Perception and Attention



Memory, Learning and Reasoning



Decision-Making and Action

Perception and Attention

Perception interprets incoming signals.

Attention selects relevant information.



Memory, Learning and Reasoning

Memory retains information.

Learning updates knowledge and skills.

Reasoning uses information to draw conclusions.



Decision-Making and Action

Decision-making selects a course of action.

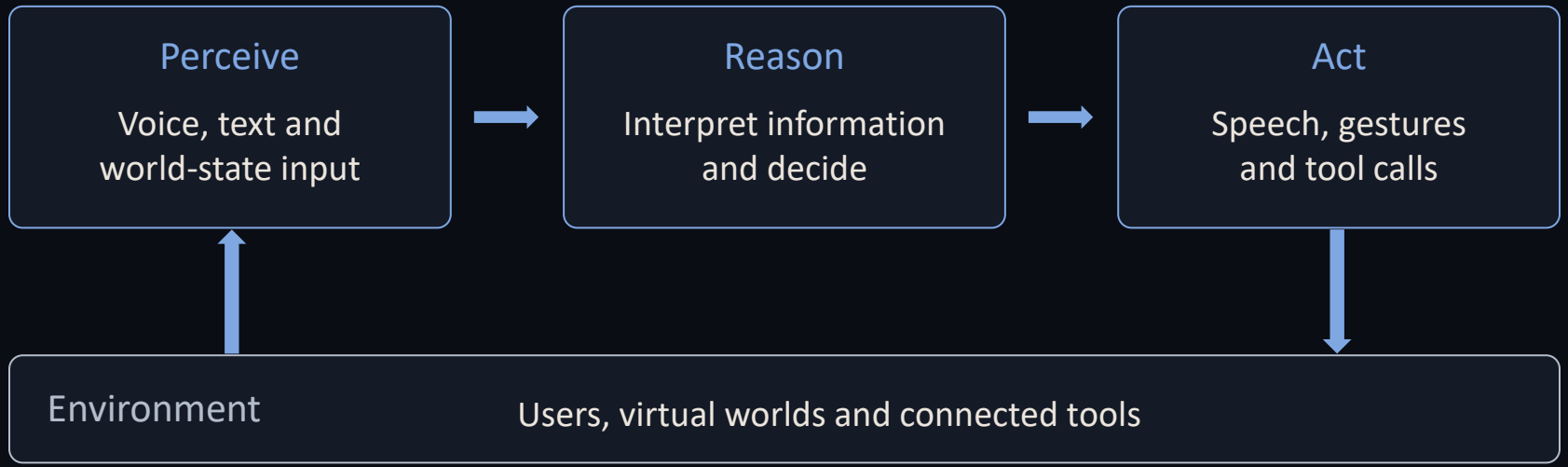
Action carries it out in the environment.



How can we organize these processes in virtual agents and humans?

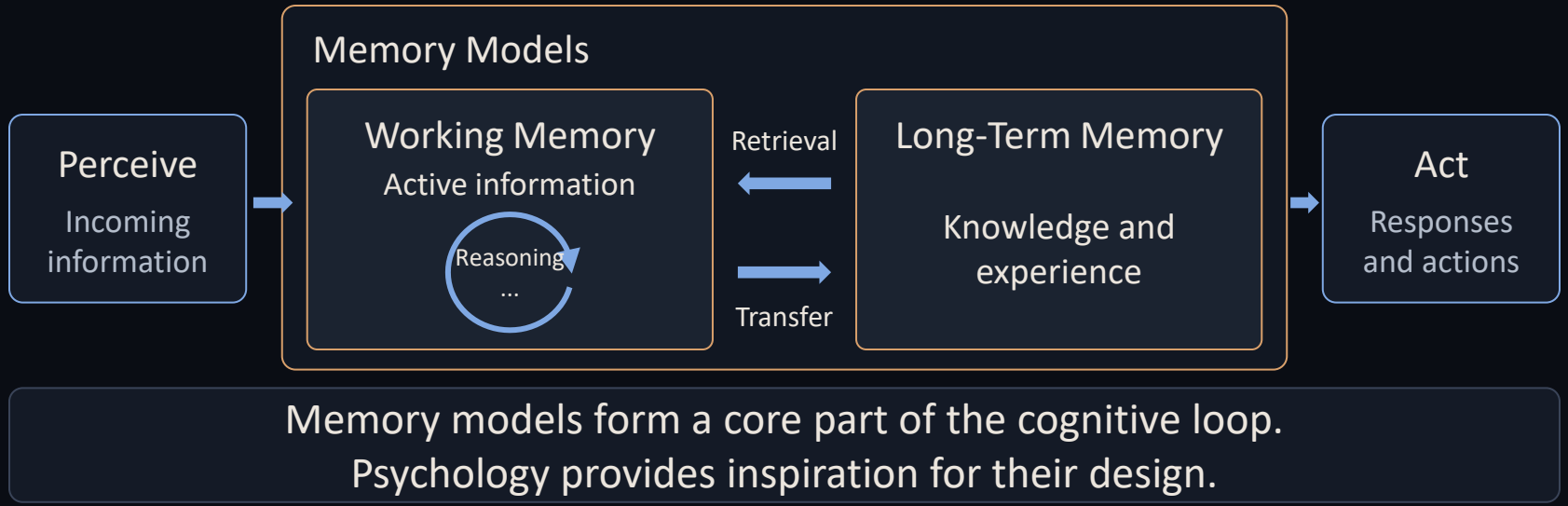
The Perceive–Reason–Act Loop

A cognitive architecture coordinates these processes over time. [2]



Memory at the Core

Reasoning draws on active information and stored knowledge. [2, 4]



Memory models form a core part of the cognitive loop.
Psychology provides inspiration for their design.

Psychological Foundations of Cognitive Architectures

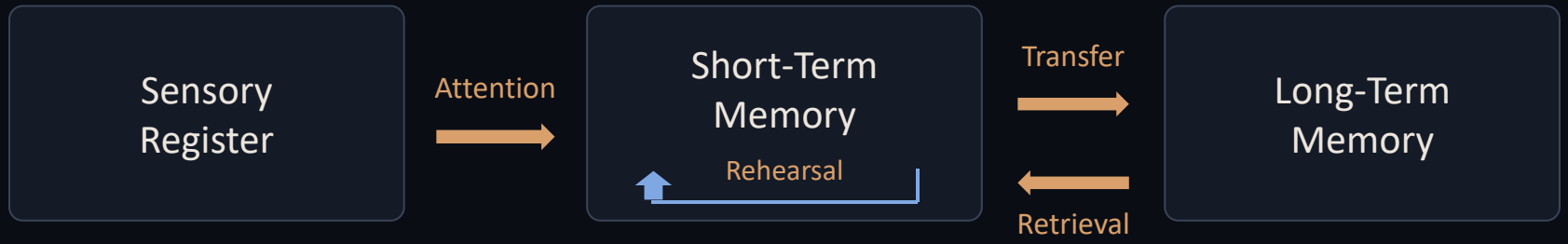
Memory Models



The Multi-Store Model [3]

Atkinson and Shiffrin describe three memory stores and their control processes. The stores differ in duration and capacity.

Memory stores



Control processes

Regulate the selection, maintenance and use of information.

Attention Rehearsal Transfer / encoding Retrieval

The Sensory Register [3]

A brief sensory trace

Briefly retains incoming sensory information before further selection and processing.



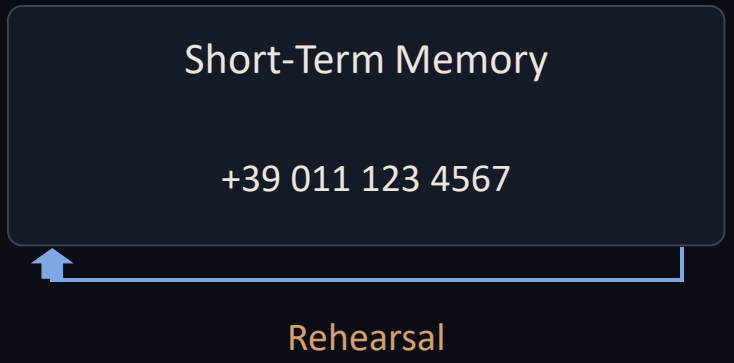
Control process

- Attention: selects information for short-term memory.

Short-Term Memory [3]

Temporary storage

A limited-capacity store that keeps information available over a short interval.



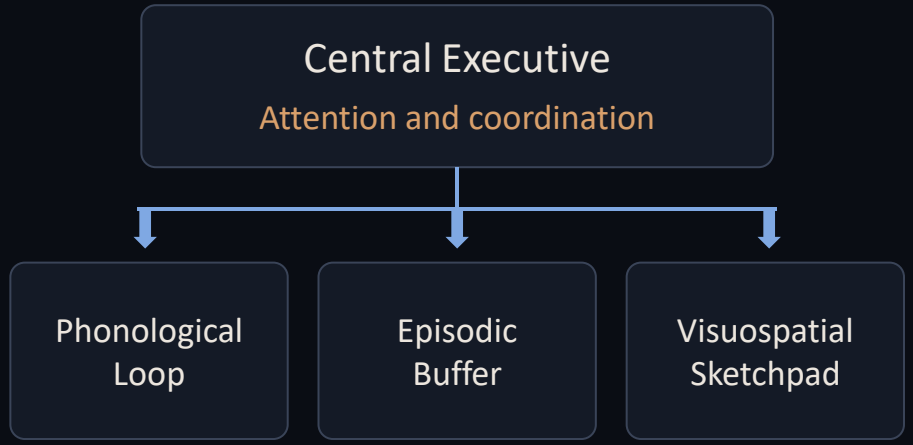
Control process

- Rehearsal: maintains information through repetition.

The Working Memory Model [4, 5]

An extension of short-term memory

Baddeley and Hitch proposed specialized stores under central executive control.



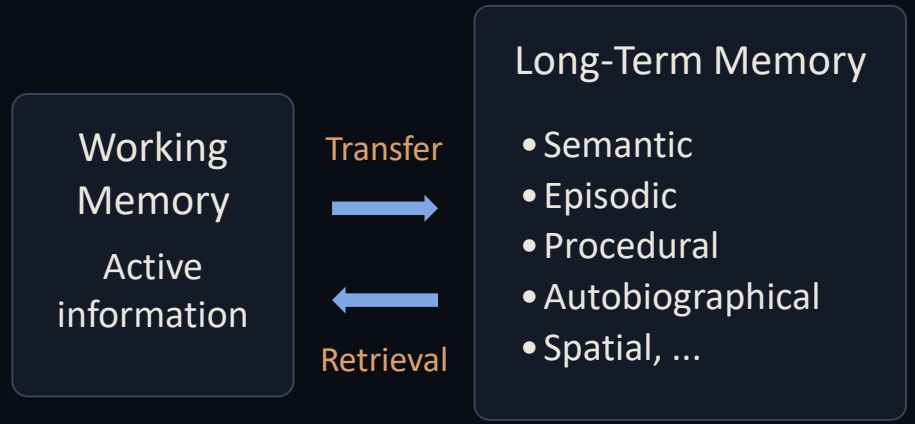
Control processes

- Attention: focuses resources on relevant information.
- Coordination: manages storage and processing across subsystems.

Long-Term Memory [3]

Retention beyond the present

Atkinson and Shiffrin describe a long-term store for knowledge and experience.



Control processes

- Transfer / encoding: supports long-term retention.
- Retrieval: brings stored information back into use.

LTM: Semantic Memory [6]

Knowledge about the world

Facts, concepts and meanings, independent of a particular personal experience.

Paris is the capital of France.



LTM: Episodic Memory [6]



Experiences in time and place

Personally experienced events, reconstructed with their context.

I arrived in Paris last summer.

LTM: Procedural Memory [7]

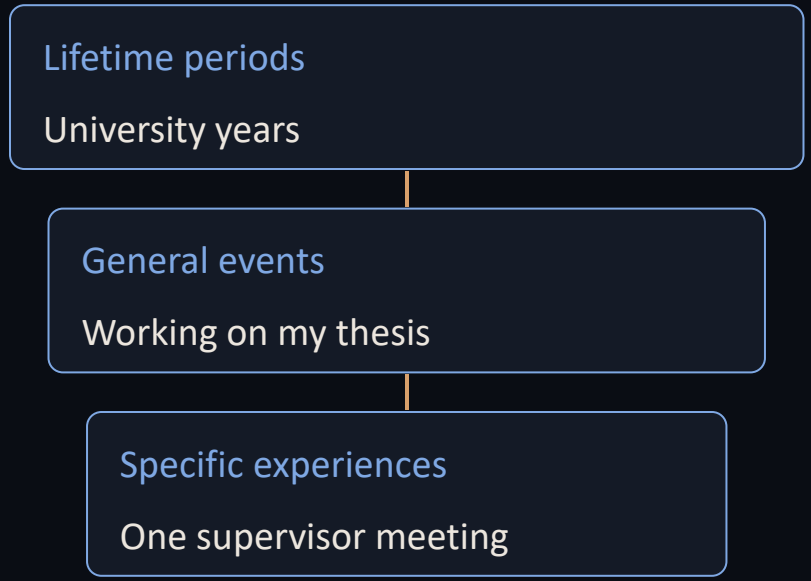
Knowing how to do something

Learned skills and habits that become fluent with practice.

Riding a bicycle.



LTM: Autobiographical Memory [8]



Experiences become a life story

Builds on episodic memory, connecting experiences with personal knowledge and identity.

My university years shaped who I am.

LTM: Spatial Memory [2, 9]

Remembering what is where

Cognitive maps support object location, navigation and spatial reasoning.

Egocentric: relative to me

The keys are to my right.

Allocentric: relative to the room

The keys are on the side table.

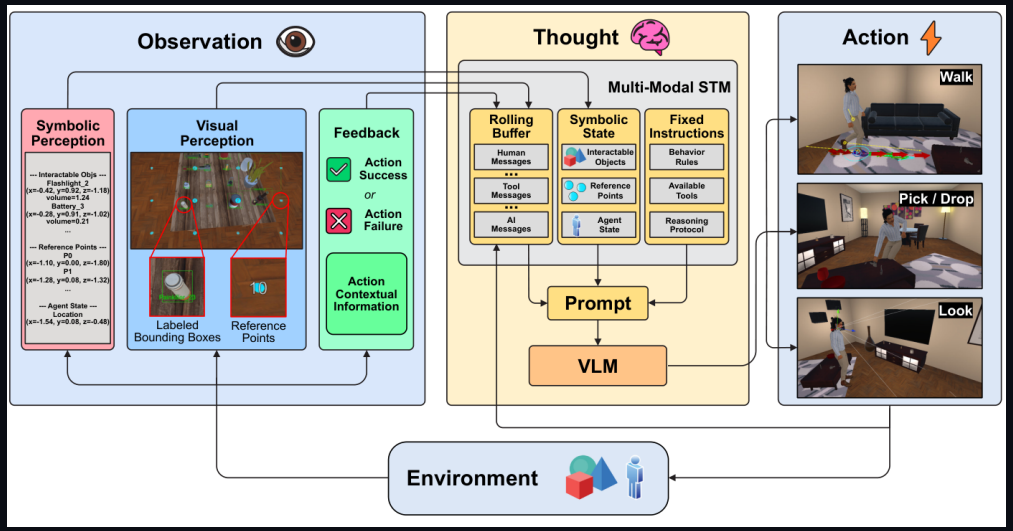


Some examples of virtual agents with memory



ADAMO: Cognitive Architecture [12]

A virtual human that follows language instructions.



Observation / Perceive

Visual + symbolic input

Thought / Reason

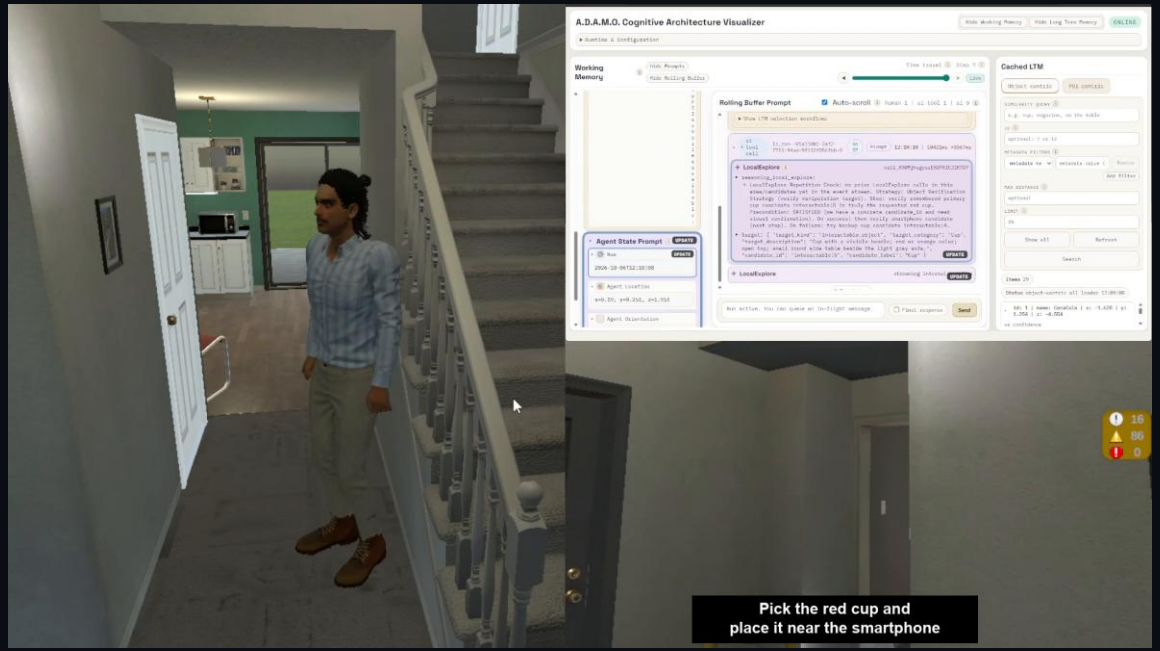
VLM + short-term memory

Action / Act

Walk, Look, Pick / Drop

Feedback starts the next cycle.

ADAMO: Virtual Human Demo [12]



Generative Agents (2023) [10]

A simulated town with 25 social agents

Agents with different identities live in Smallville, interact and plan their daily activities.

- Record events and conversations **in the memory stream**.
- Retrieve relevant memories.
- Use an LLM for planning, replies and reflection.



Generative Agents: Memory and Reflection [10]

Experiences become reusable abstract memories

Experience records

Events, conversations, activities and encounters

→

Reflection

Find patterns in important experiences

→

Abstract memories

Insights about people and their interests

Memory stream

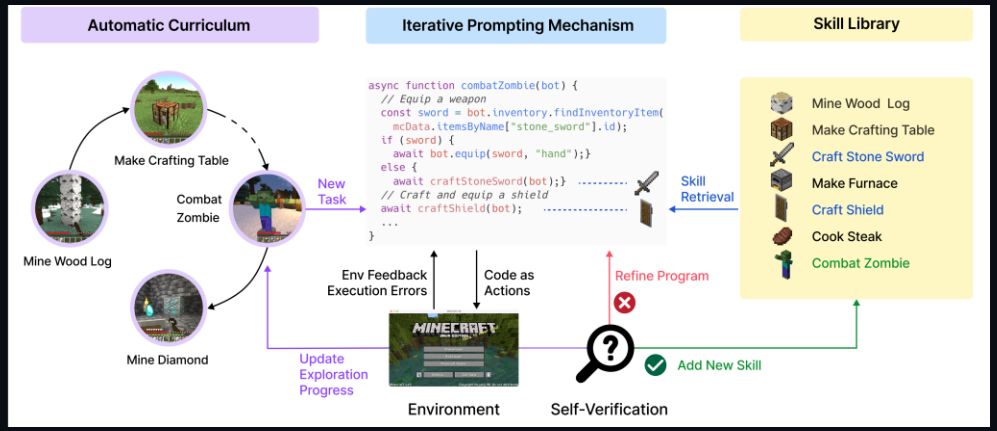
Experience	Sam helps me study and listens to my worries.
Reflection	Sam is a supportive friend.

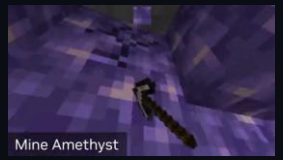
Later: I ask Sam for advice before an exam.

Reflection: a control process

Voyager: Procedural Memory [11]

Voyager is a general-purpose agent that explores and learns skills in Minecraft.





Mining



Building



Fishing

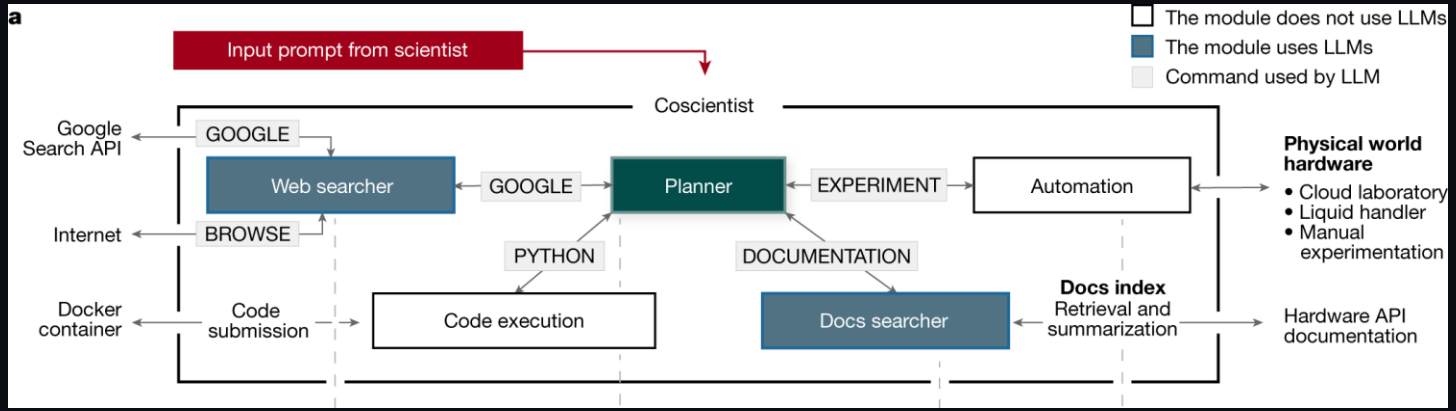


Combat

It learns, **accumulates** and **refines** skills over time.

Coscientist: Automated Chemistry [13]

Coscientist is an LLM-based research agent that plans and runs chemistry experiments.



Measured results help it choose the next experiment.

Other Agents and Applications



Project Sid [14]

Social simulation



SIMA 2 [15]

Video games



OpenVLA [16]

Real-world robotics



Project Lazarus

Healthcare training in VR

References

[1] Franklin & Graesser (1996)
Is it an Agent, or just a Program?

[2] Pecora, Strada & Bottino (2026)
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[3] Atkinson & Shiffrin (1968)
Human Memory: System and Control Processes

[4] Baddeley & Hitch (1974)
Working Memory

[5] Baddeley (2000)
The episodic buffer

[6] Tulving (1972)
Episodic and semantic memory

[7] Tulving (1985)
How many memory systems are there?

[8] Conway & Pleydell-Pearce (2000)
Autobiographical memory and the self-memory system

[9] Tolman (1948)
Cognitive maps in rats and men

[10] Park et al. (2023)
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[11] Wang et al. (2023)
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[12] Pecora et al. (2026)
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[14] Altera.AL et al. (2024)
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[15] SIMA Team (2025)
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[16] Kim et al. (2024)
OpenVLA: An Open-Source Vision-Language-Action Model